In [1]:
```
class Person:
    def __init__(gender,age,name,height, weight,blood_gp):
        self.gender = gender
        self.age = age
        self.name = name
        self.height = height
        self.weight = weight
        self.blood_gp = blood_gp
    def eat(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
    def sleep(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
    def walk(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
```

In [2]:
```
p1 = Person('Male',20,'Ghaffar', 5.8, 70, 'O+')
```

In [4]:
```
p2 = Person('Female',23,'Sara', 5.8, 60, 'O+')
p1 = Person('Male',20,'Ghaffar', 5.8, 70, 'O+')
```

In [5]:
```
p1.name
```

Out[5]:
```
'Ghaffar'
```

In [6]:
```
p2.name
```

Out[6]:
```
'Sara'
```

In [7]:
```
p1.blood_gp
```

Out[7]:
```
'O+'
```

In [8]:
```
p2.blood_gp
```

Out[8]:
```
'O+'
```

In [9]:
```
p2.eat()
```

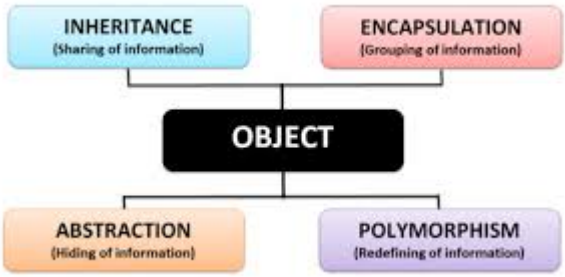
```
Sara a Female with height 5.8 and with weight 60 is eating food
```

In [10]:
```
p1.eat()
```

```
Ghaffar a Male with height 5.8 and with weight 70 is eating food
```

Four Pillars of Object Oriented Programming

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction



Encapsulation:

- Binding data with methods

In [11]:
```
p1.blood_gp
```

Out[11]: 'O+'

In [12]:
```
p1.blood_gp = "C++"
```

In [13]:
```
p1.age
```

Out[13]: 20

In [14]:
```
p2.age
```

Out[14]: 23

In [15]:
```
p2.age=119
```

In [24]:
```
class Person:
    def __init__(self,gender,age,name,height, weight,blood_gp):
        # public (can be accessed anywhere)
        self.gender = gender
        self.age = age
        self.name = name
        self.height = height
        self.weight = weight
        self._blood_gp = blood_gp
    def eat(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
    def sleep(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
    def walk(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
    def get_blood_group(self):
        return self._blood_gp
    def set_blood_group(self,new_blood_group):
        blood_groups = ['A+','A-','B+','B-','O+','O-','AB+']
        if new_blood_group in blood_groups:
            self._blood_gp = new_blood_group
        else:
            return "Invalid Blood group"
```

In [26]:
```
p2 = Person('Female',23,'Sara', 5.7, 60, 'B+')
p1 = Person('Male',20,'Ghaffar', 5.2, 70, 'O-')
```

Access Modifiers:

1. PublicScope:

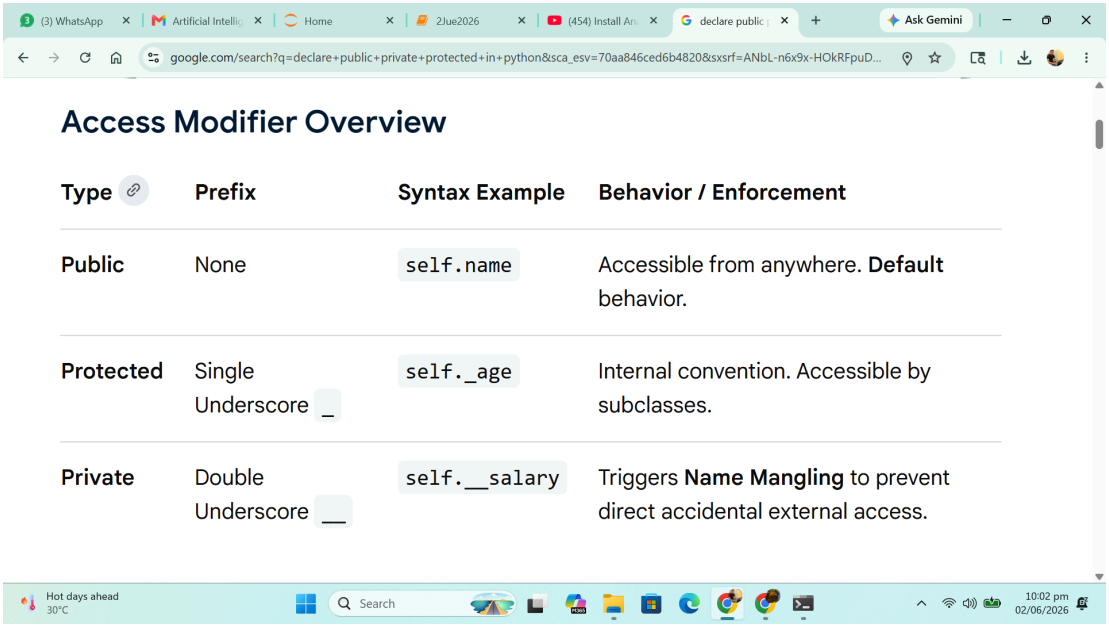
Accessible from absolutely anywhere in the program.
Purpose: Used for the main interface (API) of your class. It exposes methods that outside classes need to call to trigger actions or request safe information.
Example: A banking app exposing a public deposit() method.

2. PrivateScope:

Accessible only within the exact class where it is declared.
Purpose: This is the most restrictive level and the default setting for many languages. It isolates sensitive fields and complex internal utility functions from outside interference.
Example: A banking app setting the user's accountBalance variable to private so it cannot be directly altered from the outside without validation.

3. ProtectedScope:

Accessible within the defining class and by its derived (child) classes.
Purpose: Designed specifically for inheritance. It allows child classes to directly reuse or extend internal attributes while still hiding that data from the rest of the application.Example: A parent class Vehicle has a protected engineType field that a child class ElectricCar can modify.



In [29]:
```
p1.get_blood_group()
```

Out[29]: 'O-'

In [30]:
```
p1.set_blood_group('C++')
```

Out[30]: 'Invalid Blood group'

In [31]:
```
p1.set_blood_group('A+')
```

In [32]:
```
p1.get_blood_group()
```

Out[32]: 'A+'

Inheritance:

In [33]:
```
class Doctor:
    def __init__(self,gender,name, age, weight,height, blood_group,speciality,ho
        self.gender = gender
        self.age = age
        self.name = name
        self.height = height
        self.weight = weight
        self._blood_gp = blood_gp
        self.speciality = speciality
        self.hospital = hospital
        self.fee = fee
    def eat(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
    def sleep(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
    def walk(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
    def get_blood_group(self):
        return self._blood_gp
    def set_blood_group(self,new_blood_group):
        blood_groups = ['A+','A-','B+','B-','O+','O-','AB+']
        if new_blood_group in blood_groups:
            self._blood_gp = new_blood_group
        else:
            return "Invalid Blood group"
    def cure(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
```

In [42]:
```
class Doctor(Person):
    def __init__(self,gender,name, age, weight,height, blood_group,speciality,ho
        self.speciality = speciality
        self.hospital = hospital
        self.fee = fee
        super().__init__(gender,name, age, weight,height, blood_group)
    def cure(self):
        print(f"{self.name} a {self.gender} with height {self.height} and with w
```

In [43]:
```
d1 =Doctor('Female',23,'Sara', 5.7, 60, 'B+', 'eye','abc',1000)
```

In [38]:
```
class Student:
    def __init__(self):
        pass
    def __init__(self,name):
        self.name =name
    pass
```

In [39]:
```
s1 = Student()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[39], line 1
----> 1 s1 = Student()

TypeError: Student.__init__() missing 1 required positional argument: 'name'
```

IS A relationship:

Doctor IS A person
Student IS A person
Dog IS an Animal
NeemTree is A Tree

Object as Attribute

In [ ]:
```

```